



## Nervous Coordination, Senses and Homeostasis

Name: \_\_\_\_\_

Class: \_\_\_\_\_

1. Which of the following correctly pairs a stimulus with the type of receptor that detects it in humans?

- A. Sound waves – thermoreceptors in the skin
- B. Light – photoreceptors in the retina
- C. Blood glucose concentration – photoreceptors
- D. Touch – chemoreceptors on the tongue

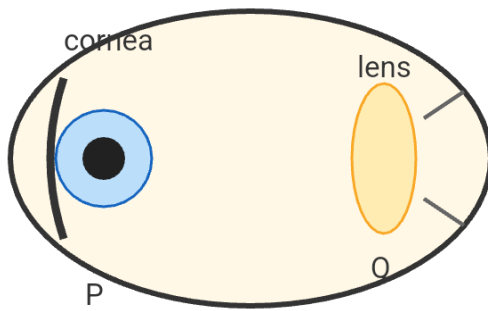
2. Which of the following structures belongs to the central nervous system?

- A. Optic nerve
- B. Retina
- C. Spinal cord
- D. Sensory nerve endings in the skin

3. During a voluntary action such as writing Chinese characters neatly, which part of the brain fine-tunes the muscular movements so that they are smooth and accurate?

- A. Medulla oblongata
- B. Cerebellum
- C. Hypothalamus
- D. Pituitary gland

4. The diagram shows a simplified section of the human eyeball.



When a student looks from a distant MTR sign to a close smartphone screen, which change correctly describes structures P and Q?

- A. P contracts; Q becomes thicker
- B. P contracts; Q becomes thinner
- C. P relaxes; Q becomes thicker
- D. P relaxes; Q becomes thinner

5. Which statement about rod cells and cone cells is correct?

- A. Rod cells are densely packed at the yellow spot and detect colour.
- B. Cone cells work best in dim light and detect black and white only.
- C. Cone cells are concentrated at the yellow spot and enable colour vision in bright light.
- D. Rod cells and cone cells are equally distributed across the whole retina.

6. A student cannot see distant objects clearly. Light rays from a distant lamp are focused in front of the retina. Which combination correctly identifies the eye defect and the corrective lens?

- A. Long-sightedness; convex lens
- B. Long-sightedness; concave lens
- C. Short-sightedness; convex lens
- D. Short-sightedness; concave lens

7. Which structure of the ear equalises air pressure on both sides of the eardrum?

- A. Cochlea
- B. Eustachian tube
- C. Semicircular canals
- D. Ear bones

8. What is the correct sequence of structures through which vibrations pass after sound waves reach the eardrum?

- A. Ear bones → oval window → cochlea
- B. Cochlea → ear bones → oval window
- C. Oval window → cochlea → ear bones
- D. Ear bones → cochlea → oval window

9. Several hours after a meal, blood glucose concentration begins to fall. Which change helps restore the normal blood glucose level?

- A. Increased secretion of insulin, converting glucose to glycogen in the liver
- B. Increased secretion of glucagon, converting glycogen to glucose in the liver
- C. Decreased secretion of glucagon, converting glucose to fat
- D. Increased uptake of glucose by muscle cells only, with no liver involvement

10. Which comparison between Type 1 and Type 2 diabetes mellitus is correct?

- A. Type 1: cells resistant to insulin; Type 2: little or no insulin produced
- B. Type 1: little or no insulin produced; Type 2: cells less sensitive to insulin
- C. Type 1: controlled by diet alone; Type 2: always requires insulin injections from onset
- D. Type 1: mainly caused by obesity; Type 2: mainly an autoimmune destruction of pancreas

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## Answers

1 B

Receptors are specialised structures that detect specific environmental changes called stimuli.

Photoreceptors in the retina detect light. Thermoreceptors detect temperature changes. Mechanoreceptors in the ear and skin detect sound vibrations or pressure. Chemoreceptors detect chemicals (taste and smell).

Therefore the correct pairing is light detected by photoreceptors in the retina.

2 C

The central nervous system (CNS) consists of the brain and the spinal cord only.

The peripheral nervous system includes cranial and spinal nerves and sense organs such as the eye and ear. The spinal cord is therefore part of the CNS, whereas sensory receptors and peripheral nerves are not.

3 B

The cerebellum coordinates voluntary muscular movements, posture and balance so that actions are smooth and precise.

The cerebrum (motor area) initiates voluntary movements, but the cerebellum adjusts them. The medulla oblongata controls involuntary actions such as breathing and heart rate. The hypothalamus is mainly concerned with homeostasis.

4 A

Structure P is the ciliary muscle region and structure Q is the lens.

For a nearby object, ciliary muscles contract. This allows the suspensory ligaments to slacken, so the elastic lens becomes thicker (more convex) and increases its refractive power. The image can then be focused on the retina.

5 C

Cone cells are concentrated at the yellow spot (fovea). They function in bright light, provide colour vision and give high visual acuity.

Rod cells are more numerous in the peripheral retina. They function in dim light, detect black and white only, and give lower acuity.

Therefore cones are responsible for detailed colour vision in daylight, not rods.

6 D

When the image of a distant object forms in front of the retina, the person has short-sightedness (myopia). This is often due to an eyeball that is too long or a lens that is too thick.

A concave lens diverges the incoming light rays slightly so that the eye can focus the image exactly on the retina.

7 B

The Eustachian tube connects the middle ear cavity to the pharynx. When it opens (for example during swallowing or yawning), air can enter or leave the middle ear so that pressure on both sides of the eardrum becomes equal.

Equal pressure allows the eardrum to vibrate freely. The cochlea converts vibrations into nerve impulses, the ear bones transmit vibrations, and the semicircular canals are concerned with balance.

8 A

Sound waves make the eardrum vibrate. The vibrations are then transmitted and amplified by the three ear bones (ossicles). The last ear bone pushes on the oval window, which sets up pressure waves in the fluid of the cochlea. Hair cells in the cochlea are stimulated and generate nerve impulses.

Therefore the order after the eardrum is: ear bones → oval window → cochlea.

9

B

When blood glucose falls below the set point, the pancreas secretes glucagon.

Glucagon acts on the liver to stimulate the conversion of stored glycogen into glucose, which is released into the blood. This raises blood glucose back toward normal (negative feedback).

Insulin has the opposite effect: it promotes conversion of glucose to glycogen when blood glucose is high.

10

B

Type 1 diabetes: the pancreas produces little or no insulin, so patients usually need insulin injections.

Type 2 diabetes: body cells become less sensitive to insulin (insulin resistance). It is often linked to lifestyle and may initially be managed by diet and exercise.

The statements that reverse these causes or treatments are incorrect.